Timer 2 Initialization Overview

Overview & Implementation Steps

Timer 2 Initialization Overview

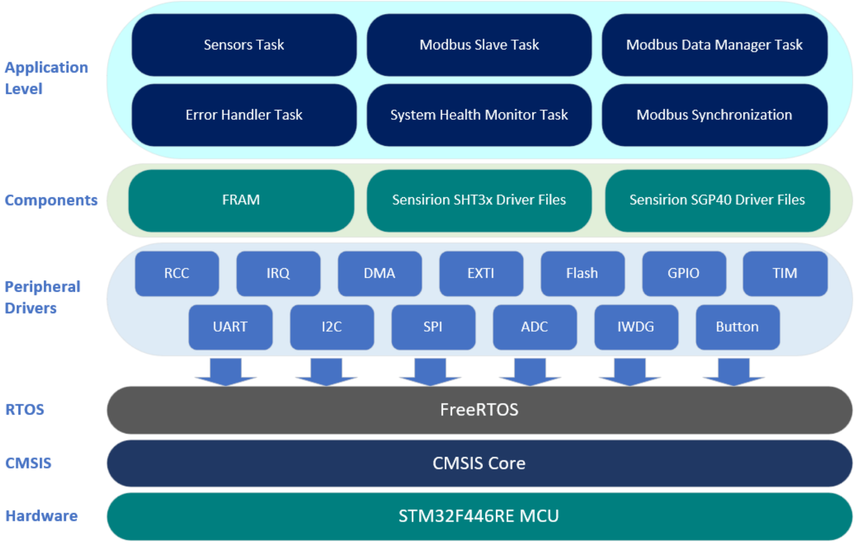
- Timers Used in the Project
- TIM2 Initialization Function

Timers Used in the Project

Timers General

- Purpose & Functionality: Crucial for generating precise delays, triggering periodic events, measuring time intervals and ensure synchronized operations across various tasks and peripherals.

- Project Significance:
  - **TIM2**: Triggers ADC1 conversions for the internal temperature sensor at 1 Hz.
  - **TIM3**: Generates high-frequency interrupts for FreeRTOS runtime statistics.
  - **TIM5**: Creates microsecond delays, required by the Sensirion drivers.



TIM2 Initialization Function

Overview: Timer 2 is a general-purpose 32-bit timer configured to trigger ADC1 conversions at a specific frequency.

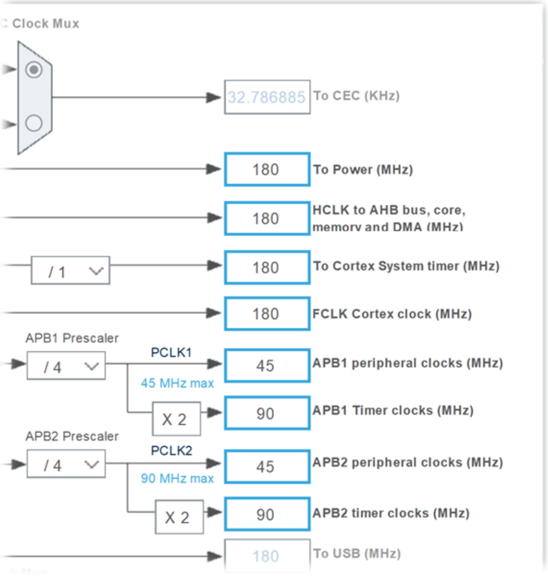
- ✓ **TIM2 Clock Enable**: Using the RCC_TIM2_CLK_ENABLE () macro, enable the TIM2 clock connected to the APB1 bus.
- ✓ **Prescaler & ARR:**
  - ✓ **Prescaler:** Configures the timer frequency; APB1 clock at 90 MHz and a prescaler (PSC) of 8999, the timer frequency is 10 KHz.

$$Timer\ frequency = \ 90\ MHz/(1 + 8999)\ = 10\ KHz$$

  - ✓ **ARR:** Sets the trigger frequency to 1 Hz.

$$Trigger\ frequency = 10{,}000\ Hz/(1 + 9999) = 1\ Hz$$

- ✓ **Master Mode Selection**: Set the update event as the trigger output (TRGO) by configuring the MMS bits in CR2.
- ✓ **Event Generation**: Set the UG bit to generate an update event.
- ✓ **Counter Enable**: Enable the TIM2 counter to start the timer.



- Project Significance: Completing the timer setup for ADC1 trigger ensures the ADC performs conversions based on the external trigger from TIM2.

Next: TIM Header File